CS220 Design and Analysis of Algorithms

School of Computer Science and Engineering
Macau University of Science and Technology
Assignment 2
Due Day: 23 April, 2026 (Thursday)

1. Show that, with the array representation for storing an n-element heap, the leaves are the nodes indexed by $\lfloor n/2 \rfloor + 1, \lfloor n/2 \rfloor + 2, \ldots, n$.

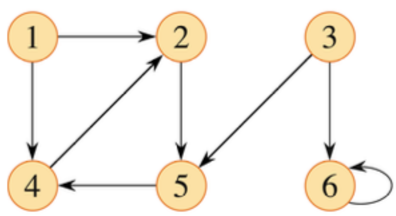
2. Show that each child of the root of a n-node heap is the root of a subtree containing at most $2n/3$ nodes.

3. Follow the figures of the example on slide 37 of Lecture 8's notes, illustrate the operation of **HEAPSORT** on the array A with numbers 5, 13, 2, 25, 7, 17, 20, 8, 4.

4. Determine the cost and structure of an **optimal binary search tree** for a set of n = 7 keys with the following probabilities. **You have to provide the *w* (sum of all the probabilities in the subtrees), *e* (expected search cost), and *root* (root) tables as well as the resulting optimal binary search tree**.

i	0	1	2	3	4	5	6	7
p_i		0.04	0.06	0.08	0.02	0.10	0.12	0.14
q_i	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05

5. Show the ***d* (distance) and *π* (predecessor)** values and the **queue Q** that results from running **breadth-first search** on the directed graph of the following figure, using vertex **3** as the source.



6. Show how the depth-first search works on the following figure. Assume that the **DFS** procedure considers the vertices in **alphabetical** order, and assume that each adjacency list is ordered **alphabetically**. Show the **discovery and finish times** for each vertex.